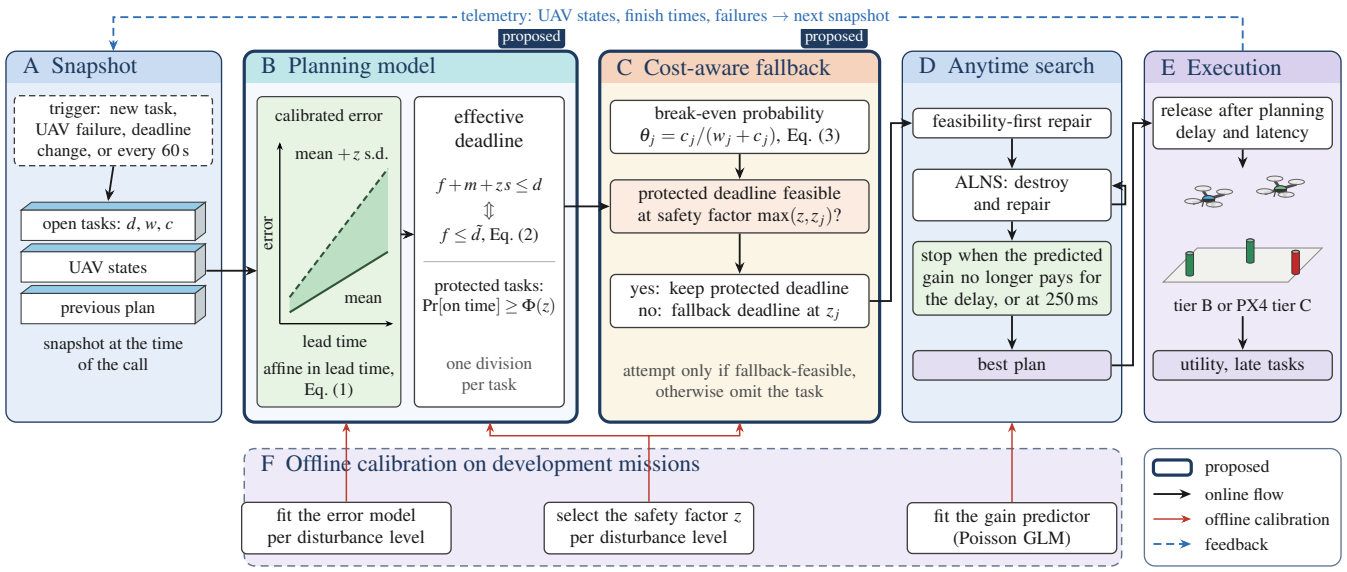

telemetry: UAV states, finish times, failures → next snapshot
A Snapshot
trigger: new task, UAV failure, deadline change, or every 60 s
open tasks: d, w, c
UAV states
previous plan
snapshot at the time of the call
proposed
B Planning model
calibrated error
mean + z s.d.
error
mean
lead time
affine in lead time, Eq. (1)
effective deadline
f + m + z s ≤ d
⇕
f ≤ d̃, Eq. (2)
protected tasks: Pr[on time] ≥ Φ(z)
one division per task
proposed
C Cost-aware fallback
break-even probability θ_j = c_j/(w_j + c_j), Eq. (3)
protected deadline feasible at safety factor max(z, z_j)?
yes: keep protected deadline
no: fallback deadline at z_j
attempt only if fallback-feasible, otherwise omit the task
D Anytime search
feasibility-first repair
ALNS: destroy and repair
stop when the predicted gain no longer pays for the delay, or at 250 ms
best plan
E Execution
release after planning delay and latency
tier B or PX4 tier C
utility, late tasks
F Offline calibration on development missions
fit the error model per disturbance level
select the safety factor z per disturbance level
fit the gain predictor (Poisson GLM)
proposed
online flow
offline calibration
feedback